

Quantum Mechanics III

Homework 8

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Due 2026-10-25 18:00

Topic

Midterm Review

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Midterm Review. State all assumptions and conventions.

$$P \rightarrow \rho \rightarrow \mathcal{E} \rightarrow \mathcal{M}_k \rightarrow p(k)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$CP : J_{\mathcal{E}} \geq 0$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$\text{counts} \sim \text{Multinomial}(N, C_{\text{det}} p)$$

Show enough intermediate work that another student can reproduce the calculation.